

DEVELOPMENT OF WEB USER INTERFACE

HTML Resource: <https://bit.ly/2So6vhq>

1. HTML

1.1 Introduction to HTML

1.1.1 What is HTML?

Hypertext Markup Language (HTML) is the standard markup language for creating web pages and web applications. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document.

Hypertext means that the document contains links that allow the reader to jump to other places in the document or to another document altogether. The latest version is known as HTML5. **A Markup Language** is a way that computers speak to each other to control how text is processed and presented.

A web page (also written as webpage) is a document that is suitable to act as a web resource on the World Wide Web. When accessed by a web browser it may be displayed as a web page on a monitor or mobile device. HTML describes the structure of Web pages using markup.

- HTML elements are the building blocks of HTML pages
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on
- Browsers do not display the HTML tags, but use them to render the content of the page

1.1.2 HTML Editors

To write HTML, we must ensure that we have the right tools. Most important, we need an HTML editor and browser to display the web page.

Plain Text Editors

Web pages can be created and modified by using professional HTML editors. However, for learning HTML we recommend starting with plain text editors like Notepad (Windows) or TextEdit (Linux). Using a simple text editor is a good way to learn HTML.

Professional HTML Editors:

There are many choices, and Here are a handful of the most popular:

Sublime Text 3: it is free and also offers cross-platform support for Windows, Mac, and Linux users.

Notepad ++: It is a tiny program to download and perform the functions you need for writing clean code.

Visual studio code: Visual studio code has syntax highlighting, smart completions and customizable formatting. It offers up suggestions for elements, tags, some values (as defined in HTML5),

Activity1 .0 : Follow the steps below to create your first web page with Notepad or TextEdit.

1. Using your computer, open notepad and type the following example of html document.
2. Save the file under *rca.htm*, then *rca.html*
3. Open the file using a browser.

```
<!DOCTYPE html>
<html>
<head>
<title>Rwanda Coding Academy</title>
</head>
<body>

<h1>Welcome to Rwanda Coding Academy</h1>
<p> First paragraph</p>

</body>
</html>
```

1.2 The HTML SYNTAX

HTML uses tags for its syntax to define HTML elements. An HTML element usually is indicated by a **start** tag and **end** tag, with the content inserted in between.

ELEMENT Syntax: <tagname>My content </tagname>

- HTML elements are not case-sensitive
- HTML elements can have attributes and content
- HTML elements usually have start tags and end tags
- If an element contains text content or nested elements, then it needs both a start and an end tag
- The entire document is placed within an *html* element
- Some elements don't require end tags, e.g. <hr>,

1.2.1 The basic syntax of HTML page document

```
<!DOCTYPE html>
<html>
<head>
<title></title>
</head>
<body>

</body>
</html>
```

1.2.2 The components of a web page in the above syntax are explained below:

1. The `<! DOCTYPE html>` declaration defines this document to be HTML5
2. The `<html>` element is the root element of an HTML page
3. The HTML **<head>** Element

The `<head>` element is a container for metadata (data about data) and is placed between the `<html>` tag and the `<body>` tag. HTML metadata is data about the HTML document. Metadata is not displayed.

Metadata typically define the document title, character set, styles, links, scripts, and other meta information. The following tags describe metadata: `<title>`, `<style>`, `<meta>`, `<link>`, `<script>`, and `<base>`

4. The `<title>` element specifies a title for the document

The `<title>` element defines the title of the document, and is required in all HTML/XHTML documents. The `<title>` element:

- a. Defines a title in the browser tab
- b. Provides a title for the page when it is added to favorites

- c. Displays a title for the page in search engine results
- 5. The `<body>` element contains the visible page content. The `<body>` element contains all the contents of an HTML document, such as text, hyperlinks, images, tables, lists, etc.

1.2.3 HTML EXTENSIONS

An html document must be saved with **.html** or **.htm** extension to be recognized by web browser. HTML is a file extension used interchangeably with HTML.

1.3. HTML ELEMENTS

1.3.1. Types of HTML Elements

Every HTML element has a default display value. The default display value for most elements is block or inline.

1.3.1.1 Block-level Elements

A block-level element **always starts on a new line** and takes up the **full width** available (stretches out to the left and right as far as it can).

Block level elements in HTML:

Block level elements	Description
<code><address></code>	Define address in html
<code><article></code>	Define an article in html
<code><blockquote></code>	Define a quote from a given source.
<code><dd></code>	The <code><dd></code> tag is used to describe a term/name in a description list.
<code><div></code>	The <code><div></code> tag defines a division or a section in an HTML document.
<code><dl></code>	The <code><dl></code> tag defines a description list.
<code><dt></code>	The <code><dt></code> tag defines a term/name in a description list.
<code><fieldset></code>	The <code><fieldset></code> tag is used to group related elements in a form. The <code><fieldset></code> tag draws a box around the related elements.
<code><figcaption></code>	The <code><figcaption></code> tag defines a caption for a <code><figure></code> element.
<code><figure></code>	<code><figure></code> element is used to mark up a photo in a document.
<code><footer></code>	The <code><footer></code> tag defines a footer for a document or section. It may contains:

	authorship information copyright information contact information sitemap back to top links related documents
<u><form></u>	Define a form to collect user data in html
<u><h1>-<h6></u>	Headings
<u><header></u>	
<u><hr></u>	Horizontal rule which defines a thematic break in an HTML page
<u></u>	The tag defines a list item
<u><main></u>	
<u><noscript></u>	The <noscript> tag defines an alternate content for users that have disabled scripts in their browser or have a browser that doesn't support script. The <noscript> element can be used in both <head> and <body>.
<u></u>	Ordered list in html
<u><p></u>	Define a paragraph
<u><pre></u>	
<u><section></u>	The <section> tag defines sections in a document, such as chapters, headers, footers, or any other sections of the document.
<u><table></u>	Define tables in html
<u><tfoot></u>	
<u></u>	Unordered list
<u><video></u>	An html tag for including video

[<address>](#) [<article>](#) [<aside>](#) [<blockquote>](#) [<canvas>](#)
[<dd>](#) [<div>](#) [<dl>](#) [<dt>](#) [<fieldset>](#)
[<figcaption>](#) [<figure>](#) [<footer>](#) [<form>](#) [<h1>-<h6>](#)
[<header>](#) [<hr>](#) [](#) [<main>](#) [<nav>](#)
[<noscript>](#) [](#) [<p>](#) [<pre>](#) [<section>](#)
[<table>](#) [<tfoot>](#) [](#) [<video>](#)

1.3.1.2 HTML Inline Elements

Generally, inline elements may contain only data and other inline elements. You can't put block elements inside inline elements.

By default, inline elements do not force a new line to begin in the document flow. Block elements, on the other hand, typically cause a line break to occur.

Examples of Inline elements:

<a>	<abbr>	<acronym>		<bdo>	<big>	
<button>	<cite>	<code>	<dfn>		<i>	
<input>	<kbd>	<label>	<map>	<object>	<output>	<q>
<samp>	<script>	<select>	<small>			<sub>
<sup>	<textarea>	<time>	<tt>	<var>		

Name	Description
HTML
 Tag	Add a break
• HTML <button> Tag	The <button> tag in HTML is used to define the clickable button. <button> tag is used to submit the content.
• HTML <time> Tag	The <time> tag is used to display the human-readable date/time.
• HTML <a> Tag	The <a> tag (anchor tag) in HTML is used to create a hyperlink on the webpage.
• HTML <abbr> Tag	The <abbr> tag(Abbreviation) in HTML is used to define the abbreviation or short form of an element.
• HTML Tag	Illustrates highlighting the text by making it as bold text in HTML.
• HTML <cite> Tag	The <cite> tag in HTML is used to define the title of a work.
• HTML <code> Tag	The <code> tag in HTML is used to define the piece of computer code.
• HTML <dfn> Tag	The <dfn> tag in HTML represents definition element and is used to representing a defining instance in HTML.
• HTML Tag	The tag in HTML is a phrase tag and used to emphasize the text content
• HTML <i> Tag	The <i> tag in HTML is a phrase tag and used to emphasize the text content
• HTML <sub> Tag	This <sub> tag in HTML is used to write text below the baseline of the text in a smaller font.
• HTML <sup> Tag	The <sup> tag in HTML describes the text as a superscript text
HTML Tag	HTML tag is used to add image inside webpage/website.
HTML <label> Tag	The <label> tag in HTML is used to provide a usability improvement for mouse users i.e, if a user clicks on the text within the <label> element, it toggles the control. The <label> tag defines the label for <button>, <input>, <meter>, <output>, <progress>, <select>, or <textarea> element.

1.3.2 HTML ATTRIBUTES

Attributes provide additional information about HTML elements.

- All HTML elements can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**

- Attributes can be specific or global attributes

Examples:

1. The href Attribute

HTML links are defined with the `<a>` tag. The link address is specified in the `href` attribute:

```
<a href="io.html"> Input and output devices</a>
```

1.3.3 Html headings `<h1> ...<h6>`

Headings are defined with the `<h1>` to `<h6>` tags.

`<h1>` defines the most important heading. `<h6>` defines the least important heading.

Activity:

1. Add the following code to html document, save it and preview the result in the browser.

```
<h1>Heading no. 1</h1>
<h2>Heading no. 2</h2>
<h3>Heading no. 3</h3>
<h4>Heading no. 4</h4>
<h5>Heading no. 5</h5>
<h6>Heading no. 6</h6>
```

1.3.4 HTML paragraphs `<p>`

HTML paragraph or HTML p tag is used to define a paragraph in a webpage. Let's take a simple example to see how it work. It is a notable point that a browser itself add an empty line before and after a paragraph. An HTML `<p>` tag indicates starting of new paragraph.

If there are various `<p>` tags in one HTML file then browser automatically adds a single blank line between the two paragraphs.

Activity:

Add the following code to your html document, save it and preview the result in the browser.

```
<p>This is first paragraph.</p>
<p>This is second paragraph.</p>
<p>This is third paragraph.</p>
```


1.3.5 The <div> tag in html

The <div> element is a block-level element. The <div> tag defines a division or a section in an HTML document.

The <div> element is often used as a **container** for other HTML elements to style them with CSS or to perform certain tasks with JavaScript.

```
<!DOCTYPE html>
<html>
<head>
<title> Block elements</title>
</head>
<body>
<div>Rwanda Coding Academy</div>
<div>Academic Year 2019</div>

<p>The DIV element is a block element, and will start on a new line.</p>

</body>
</html>
```

1.3. 6 HTML breaks
 and Horizontal rule <hr/>

The
 tag inserts a single line break.

The
 tag is an empty tag which means that it has no end tag.

A line break is marked up as follows:

Example:

This text contains
a line break.

Answer:

This text contains

a line break.

1.3.7 HTML Horizontal rule<hr>

The <hr> tag in HTML stands for horizontal rule. The <hr> tag is an empty tag and it does not require an end tag.

The <hr> tag a thematic break in an HTML page (e.g. a shift of topic).

The <hr> element is used to separate content (or define a change) in an HTML page.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML hr tag</title>
  </head>
  <body>
    <p>There is a horizontal rule below this paragraph.</p>
    <hr>
    <p>This is a horizontal rule above this paragraph.</p>
  </body>
</html>
```

1.3.8 HTML Tables

HTML table tag is used to display data in tabular form (row * column). There can be many columns in a row. Table is defined by <table>, row is defined by <tr>, and data is defined by <td>

HTML tables are used to manage the layout of the page e.g. header section, navigation bar, body content, footer section etc. But it is recommended to use div tag over table to manage the layout of the page .

Tag	Description
<table>	It defines a table.
<tr>	It defines a row in a table.
<th>	It defines a header cell in a table.
<td>	It defines a cell in a table.
<caption>	It defines the table caption.
<colgroup>	It specifies a group of one or more columns in a table for formatting.
<col>	It is used with <colgroup> element to specify column properties for each column.
<tbody>	It is used to group the body content in a table.
<thead>	It is used to group the header content in a table.
<tfoot>	It is used to group the footer content in a table.

<colgroup>, <col>,<thead>,<tbody>,<tfooter> are used to add Cascading Style Sheet(CSS) on the table. These will be discussed in the CSS section.

Activity: Create the following table in your html document

Number	Province	District
101	Kigali city	Gasabo
102	Kigali city	Nyarugenge
103	Kigali	Kicukiro
201	Western Province	Nyabihu
202	Western Province	Rubavu

Results

```

<!DOCTYPE html>
<html>
<head>
<title>Rwanda Coding Academy</title>
</head>
<body>
<h2> List of Districts in Rwanda</h2>
<table border=1 cellspacing=0, cellpadding=3 width="400px">

<tr>
<th> Code</th>
<th> Province </th>
<th> District</th>
</tr>
<tr>
<td> 101</td>
<td>Kigali City </td>
<td> Gasabo</td>
</tr>
<tr>
<td> 102</td>
<td> Kigali City</td>
<td> Kicukiro</td>
</tr>
<tr>
<td> 103</td>
<td> Kigali City</td>
<td> Nyarugenge</td>
</tr>
<tr>
<td> 201</td>
<td> Western Province</td>
<td> Nyabihu</td>
</tr>
<tr>
<td> 202</td>
<td> Western Province</td>
<td> Rubavu</td>
</tr>
</table>
</body>
</html>

```

ROWSPAN AND COLSPAN

Sometimes it makes sense for a cell to span multiple columns or multiple rows. This might be used for a header cell that titles a group of columns, or a side-bar that groups rows of entries. **Colspan** and **rowspan** are attributes of the two table-cell elements, **<th>** and **<td>**. They provide the same functionality as “merge cell” in spreadsheet programs like Excel.

Rowspan: Allows a single table cell to span the height of more than one cell or row

Activity: Using html table build the following table

Code	Province	District
101	Kigali City	Gasabo
102		Kicukiro
103		Nyarugenge
202	Western Province	Nyabihu
201		Rubavu

Answer:

```

<table border=1 cellspacing=0, cellpadding=3 width="400px">
<caption> List of Districts in Rwanda using ROWSPAN attribute</caption>
<tr>
<th> Code</th>
<th> Province </th>
<th> District</th>
</tr>
<tr>
<td> 101</td>
<td rowspan=3>Kigali City </td>
<td> Gasabo</td>
</tr>
<tr>
<td> 102</td>
<td> Kicukiro</td>
</tr>
<tr>
<td> 103</td>
<td> Nyarugenge</td>
</tr>
<tr>
<td> 202</td>
<td rowspan=2> Western Province</td>
<td> Nyabihu</td>
</tr>
<tr>
<td> 201</td>
<td> Rubavu</td>
</tr>
</table>

```

Colspan: Allows a single table cell to span the height of more than one cell or row.

Activity: Using html table design the following table

0-15		16-25		26-35		36-45		46-55		56-above	
Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women
800	700	700	800	600	1000	900	900	1200	2500	800	800
1500		1500		1600		1800		3700		1600	

student score

School	id	Name	subject	score	Percentage
RCA	1	John	Math	5	50%
	2	Mary	Math	10	100%
	3	Clarisse	English	7	70%
	4	Kalisa	Math	8	80%
Total				30	75%

Answer:

```

<!DOCTYPE html>
<html>
<head>
<title>Rwanda Coding Academy</title>
</head>
<body>
<h2> List of Districts in Rwanda</h2>
<table border=1 cellspacing=0 cellpadding=3>
  <caption>Population distribution by age Group</caption>
  <tr>
    <th colspan="2"> 0-15</th>
    <th colspan="2"> 16-25</th>
    <th colspan="2"> 26-35</th>
    <th colspan="2"> 36-45</th>
    <th colspan="2"> 46-55</th>
    <th colspan="2"> 46-above</th>
  </tr>
  <tr>
    <th>Men</th><th>Women</th>
    <th>Men</th> <th>Women</th>
    <th>Men</th> <th>Women</th>
    <th>Men</th> <th>Women</th>
    <th>Men</th><th>Women</th>
    <th>Men</th><th>Women</th>
  </tr>
  <tr>
    <td>800</td><td>700</td>
    <td>700</td><td>800</td>
    <td>600</td> <td>1000</td>
    <td>900</td> <td>900</td>
    <td>1200</td> <td>2500</td>
    <td>800</td> <td>800</td>
  </tr>
  <tr>
    <td colspan=2>Total 1500</td><td colspan=2>1500</td>
    <td colspan=2> 1600</td><td colspan=2>1800</td>
    <td colspan=2>3700</td><td colspan=2>1600</td>
  </tr>
</table>

</body>
</html>

```


Activity: Build the following table

student score					
School	id	Name	subject	score	Percentage
RCA	1	John	Math	5	50%
	2	Mary	Math	10	100%
	3	Clarisse	English	7	70%
	4	Kalisa	Math	8	80%
Total				30	75%

Answer:

```
<!Doctype html>
<html>
<head>
<title> My score</title>
</head>
<body>
<table border=1 width=400 cellpadding=0>
<caption>Student score</caption>
<tr>
<th>School </th>
<th>ID</th>
<th>Name</th>
<th>Subject</th>
<th>Score</th>
<th>Percentage</th>
</tr>
<tr>
<td rowspan=4>RCA</td>
<td>1</td>
<td>John</td>
<td>Math</td>
<td>5</td>
<td>50%</td>
</tr><tr>
<td>2</td>
<td>Mary</td>
<td>Math</td>
<td>10</td>
<td>100%</td>
</tr><tr>
<td>3</td>
<td>Clarisse</td>
<td>English</td>
<td>7</td>
<td>70%</td>
</tr>
<tr>
<td>4</td>
<td>Kalisa</td>
<td>Math</td>
<td>8</td>
<td>80%</td>
</tr>
<tr>
<td colspan=4> Total</td>
<td>30</td>
<td>75%</td>
</tr>
</table>
</body>
</html>
```

1.3.9 HTML Lists

HTML Lists are used to specify lists of information. All lists may contain one or more list elements. There are three different types of HTML lists:

1. Ordered List or Numbered List (ol)
2. Unordered List or Bulleted List (ul)
3. Description List or Definition List (dl)

HTML Ordered List or Numbered List

In the ordered HTML lists, all the list items are marked with numbers by default. It is known as numbered list also. The ordered list starts with `<ol>` tag and the list items start with `<li>` tag.

Activity: Add the following in the html document

```
<h2>Districts</h2>

<ol>
  <li>Gasabo</li>
  <li>Nyarugenge</li>
  <li>Nyabihu</li>
  <li>Musanze</li>
</ol>
```

In the ordered list, you can define a specific number to start with

Example: Add the following to the districts.html file, save and preview the result.

```
<ol start="5">
  <li>Rubavu</li>
  <li>Rutsiro</li>
  <li>Rusizi</li>
</ol>
```

HTML Unordered List or Bulleted List

In HTML Unordered list, all the list items are marked with bullets. It is also known as bulleted list also. The Unordered list starts with `<ul>` tag and list items start with the `<li>` tag.

Activity: Add the following in the html document

```
<h2>Districts</h2>
```

```
<ul>
<li>Gasabo</li>
<li>Nyarugenge</li>
<li>Nyabihu</li>
<li>Musanze</li>
</ul>
```

HTML Description List or Definition List

It is also known as definition list where entries are listed like a dictionary or encyclopedia. The definition list is very appropriate when you want to present glossary, list of terms or other name-value list.

The definition list has three tags:

1. **<dl> tag** defines the start of the list.
2. **<dt> tag** defines a term.
3. **<dd> tag** defines the term definition (description).

Activity: Using the definition list, build the following list

Summary of programming language

```
HTML
  Markup language
PHP
  Scripting language
JavaScript
  Scripting language
Java
  Programming language and a platform
C
  General-purpose and imperative language
```

Answer:

```
<!DOCTYPE html>
<html>
<head>
<title>Programming languages</title>
</head>
<body>
<h2>Summary of programming languages</h2>
<dl>
<dt>HTML</dt>
<dd>Markup language</dd>
<dt>PHP</dt>
<dd>Scripting language</dd>
<dt>JavaScript</dt>
<dd>Scripting language</dd>
<dt>Java</dt>
<dd>Programming language and a platform</dd>
<dt>C</dt>
<dd>General-purpose and imperative language</dd>
</dl>
</body>
</html>
```

HTML NESTED LISTS

A list within another list is termed as nested list.

Activity: Develop the following list

1. Input Devices
 - Keyboard
 - Mouse
 - Touchpad
 - Light pen
2. Output devices
 - Printer
 - Monitor
 - Speaker

Answer:

```

<!DOCTYPE html>
<html>
<head>
<title>Example of input and output devices</title>
</head>
<body>
<h2>Example of input and output devices</h2>
<ol>
  <li> Input devices
    <ul>
      <li>Keyboard</li>
      <li> Mouse</li>
      <li> Light pen</li>
      <li> Touchpad</li>
    </ul>
  </li>
  <li> Input devices
    <ul>
      <li> Monitor</li>
      <li> Printer</li>
      <li> Speaker</li>
    </ul>
  </li>
</ol>
</body>
</html>

```

1.3.10 HTML LINKS

Links are found in nearly all web pages. Links allow users to click their way from page to page. HTML links are hyperlinks. You can click on a link and jump to another document.

When you move the mouse over a link, the mouse arrow will turn into a little hand.

Syntax:

```
<a href="URL">link text</a>
```

Examples

Consider the following folder structure

File is in the same folder

```

<a href="districts.html"> List of districts in Rwanda</a><br>
<a href="io.html"> Input and output devices</a>

```

File is in the subfolder

```
<a href="others/list.html" title="List in html"> List usage in html</a>
```

Link to web address

```
<a href=" https://irembo.gov.rw/rolportal/web/rol" title="Irembo portal"> Irembo</a>  
<a href=" https://news.ycombinator.com/" title=" Hacking news"> Hacker news</a>
```

Attributes of Link<a>

Attribute	Value	Description
download	filename	Specifies that the target will be downloaded when a user clicks on the hyperlink
Href	URL	Specifies the URL of the page the link goes to
hreflang	language_code	Specifies the language of the linked document
Media	media_query	Specifies what media/device the linked document is optimized for
Ping	list_of_URLs	Specifies a space-separated list of URLs to which, when the link is followed, post requests with the body ping will be sent by the browser (in the background). Typically used for tracking.
Rel	alternate author bookmark external help license next nofollow norereferrer noopener prev search tag	Specifies the relationship between the current document and the linked document
target	_blank _parent	Specifies where to open the linked document

	_self _top	
Type	media_type	Specifies the media type of the linked document

Types of URL in a link

The URL is the address of the linked resource/document. It specifies the URL of the page the link goes to

Possible values:

- An absolute URL - points to another web site (like href="http://www.example.com/about.html")
- A relative URL - points to a file within a web site (like href="/themes/about.html")

Link Target

Value	Description
_blank	Opens the linked document in a new window or tab
_self	Opens the linked document in the same frame as it was clicked (this is default)
_parent	Opens the linked document in the parent frame
_top	Opens the linked document in the full body of the window

1.3.11 HTML Images

In HTML, images are defined with the tag.

The tag is empty, it contains attributes only, and does not have a closing tag.

The src attribute specifies the URL (web address) of the image:

Example:

```

```

1. The width and height Attributes

Html elements such as Images, tables have a set of **size** attributes, which specifies the **width** and **height**.

a) width of image

It is an optional attribute which is used to specify the width to display the image. It is not recommended now. CSS will be applied in place of width attribute.

b) height of image

It is the height of the image. The HTML height attribute also supports iframe, image and object elements. It is not recommended now. CSS will be applied in place of width attribute.

Example: ``

Attributes of HTML img tag also include:

The src and alt are important attributes of HTML img tag. All attributes of HTML image tag are given below.

2. Src of image

It is a necessary attribute that describes the source or path of the image. It instructs the browser where to look for the image on the server.

The location of image may be on the same directory or another server.

3. Alt of image

The alt attribute defines an alternate text for the image, if it can't be displayed. The value of the alt attribute describe the image in words. The alt attribute is considered good for SEO prospective.

4. The title Attribute

Here, a `title` attribute is added to the `<p>` element. The value of the title attribute will be displayed as a tooltip when you mouse over the paragraph.

```
<p title="I'm a tooltip">  
This is a paragraph.  
</p>
```

1.3.12 HTML COMMENTS

HTML comments are text or code written in HTML file to give an explanation about the code, and not visible to the user. HTML comments will be ignored by the browser, so comments will not be visible on the webpage. Comments make code easy to understand and increase readability of code. Comments are also part of the code, which gives an explanation of the code.

Syntax:

```
<!-- -->  
  
<!-- Write your comments here -->
```

Comments can be used for documentation purpose, and debugging purpose:

Example:

```

<!DOCTYPE html>
<html>
<head>
  <title> Adding Comments </title>
</head>
<!-- This is body section, write code here which you want to display on web-page With red background
color-->
  <body bgcolor= "red">
    <h2>Rwanda Coding Academy</h2>
    <!-- This is image for Rwanda Coding Academy
you can see it on your web-page
of your favorite browser -->
    
  </body>
</html>

```

1.3.13 HTML FONTS

Fonts play a very important role in making a website more user friendly and increasing content readability. Font face and color depends entirely on the computer and browser that is being used to view your page but you can use HTML **** tag to add style, size, and color to the text on your website.

The font tag is having three attributes called **size**, **color**, and **face** to customize your fonts. To change any of the font attributes at any time within your webpage, simply use the tag. The text that follows will remain changed until you close with the tag. You can change one or all of the font attributes within one tag.

Font Size

You can set content font size using **size** attribute. The range of accepted values is from 1(smallest) to 7(largest). The default size of a font is 3.

Font Face

You can set font face using *face* attribute but be aware that if the user viewing the page doesn't have the font installed, they will not be able to see it. Instead user will see the default font face applicable to the user's computer.

Font Color

You can set any font color you like using *color* attribute. You can specify the color that you want by either the color name or hexadecimal code for that color.

COLORS

Colors are very important to give a good look and feel to your website. You can specify colors on page level using <body> tag or you can set colors for individual tags using **bgcolor** attribute.

HTML Color Coding Methods

There are following three different methods to set colors in your web page –

- ☐ **Color names** – You can specify color names directly like green, blue or red. Here is the list of W3C Standard 16 Colors names and it is recommended to use them.

	Black		Gray		Silver		White
	Yellow		Lime		Aqua		Fuchsia
	Red		Green		Blue		Purple
	Maroon		Olive		Navy		Teal

- ☐ **Hex codes** – A six-digit code representing the amount of red, green, and blue that makes up the color.

In HTML, a color can be specified using a hexadecimal value in the form:

#rrggbb

Where rr (red), gg (green) and bb (blue) are hexadecimal values between 00 and ff (same as decimal 0-255).

Note: See [Hexadecimal to decimal conversion](#)

For example, #ff0000 is displayed as red, because red is set to its highest value (ff) and the others are set to the lowest value (00).

Examples



- ☐ **Color decimal or percentage values** – This value is specified using the `rgb( )` property.

RGB Value

In HTML, a color can be specified as an RGB value, using this formula:

`rgb(red, green, blue)`

Each parameter (red, green, and blue) defines the intensity of the color between 0 and 255.

For example, `rgb(255, 0, 0)` is displayed as red, because red is set to its highest value (255) and the others are set to 0.

To display the color black, all color parameters must be set to 0, like this: `rgb(0, 0, 0)`.

To display the color white, all color parameters must be set to 255, like this: `rgb(255, 255, 255)`.



HSL Value

In HTML, a color can be specified using hue, saturation, and lightness (HSL) in the form:

`hsl(hue, saturation, lightness)`

Hue is a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue.

Saturation is a percentage value, 0% means a shade of gray, and 100% is the full color.

Lightness is also a percentage, 0% is black, 50% is neither light or dark, 100% is white

`hsl(0, 100%, 50%)`

`hsl(0, 80%, 50%)`

`hsl(0, 60%, 50%)`

`hsl(0, 40%, 50%)`

`hsl(0, 20%, 50%)`

`hsl(0, 0%, 50%)`

1.3.14 HTML FORMS

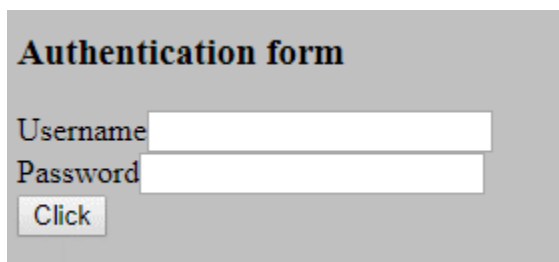
The <form> Element

- ☐ The HTML `<form>` element defines a form that is used to collect user input.
- ☐ The data is usually sent to a script on the server for processing

Example: Create an login.html file and add the following html code. Then make a preview in the browser.

```
<!DOCTYPE html>
<html>
<head>
<title>Fons</title>
</head>
<body bgcolor='silver'>
<h3>Authentication</h3>
<form action='/' method='get'>
  Username<input type='text' name='UserName' /><br>
  Password<input type='password' name='password' /><br>
  <input type='submit' value='Click' />
</form>
</body>
</html>
```

Result



The screenshot shows a web browser displaying the rendered HTML form. The form has a title 'Authentication form' in a bold, black, serif font. Below the title, there are two input fields: 'Username' and 'Password'. The 'Username' field is a standard text input, and the 'Password' field is a password input (masked with dots). Below these fields is a 'Click' button, which is a submit button with a light gray background and a thin border.

Form Elements

- The form itself is defined using a form element, with the following required attributes:
 - **action**: the name of the script on the server to send the form data to
 - **method**: how to send the data (GET or POST).
- The form will contain multiple form controls, such as input, select and textarea

The controls: input

The `<input>` element is the most important form element. The `<input>` element can be displayed in several ways, depending on the **type** attribute.

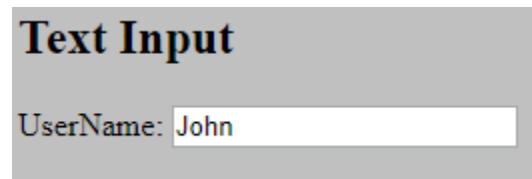
- ☐ This element takes many forms depending on the value of the type attribute
- ☐ Every input should have a **unique** name which becomes the GET or POST variable name
- ☐ The value attribute is the initial value of the input

```
<input type="text" name="firstName" value="John">
```

Input: text

- ☐ This is a single line text input
- ☐ Useful attributes:
 - **size**: the width of the text box in characters
 - **maxlength**: the maximum number of characters that can be typed into the text box

```
<input type="text" name="firstName" value="John">
```

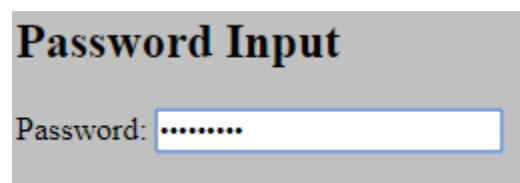


A screenshot of a web form titled "Text Input". It features a label "UserName:" followed by a single-line text input field containing the text "John".

Input: password

- ☐ Exactly the same as the previous single line textbox, except characters are masked
- ☐ Useful attributes:
 - **size**: the width of the text box in characters
 - **maxlength**: the maximum number of characters that can be typed into the text box

```
<input type="password" name="firstName">
```

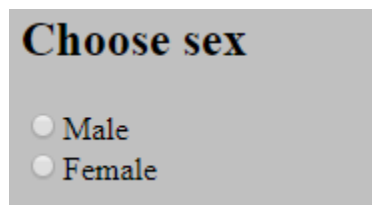


A screenshot of a web form titled "Password Input". It features a label "Password:" followed by a single-line password input field containing seven dots, indicating masked characters.

Input: radio

- ❑ These are usually used to allow the user to select **one** option from many
- ❑ When multiple radio buttons have the same name, the user can only select one at a time
- ❑ Useful attributes:
 - ❑ **checked**: set if the radiobutton should be initially selected (i.e. checked="checked")

```
<!DOCTYPE html>
<html>
<head>
<title>Form</title>
</head>
<body bgcolor='silver'>
<h2>Choose sex</h2>
<form action='/' method='post'>
<input type='radio' name='sex' value='male'>Male<br>
<input type='radio' name='sex' value='female'>Female<br>
</form>
</body>
</html>
```



The screenshot shows a web browser window with a title bar. Inside, there's a form with a heading 'Choose sex' in a large, bold, serif font. Below the heading, there are two radio buttons. The first is labeled 'Male' and the second is labeled 'Female'. Both radio buttons are currently unselected.

Inputs: checkbox

- ❑ These are used to allow the user to select Zero or More options
- ❑ Useful attributes:
 - ❑ **checked**: set if the checkbox should be initially selected (i.e. checked="checked")

```
<input type='checkbox' name='Kinyarwanda' value='Kinyarwanda'>Kinyarwanda<br>
<input type='checkbox' name='English' value='English'>English<br>
<input type='checkbox' name='French' value='French'>French<br>
<input type='checkbox' name='Kiswahili' value='Kiswahili'>Kiswahili<br>
```

Spoken Languages

- ☐ Kinyarwanda
- ☐ English
- ☐ French
- ☐ Kiswahili

Input: Submit

`<input type="submit">` defines a button for **submitting** form data to a **form-handler**.

The form-handler is typically a server page with a script for processing input data. The form-handler is specified in the form's `action` attribute:

```
<!DOCTYPE html>
<html>
<body bgcolor='silver'>
<h2>Submit Button</h2>
<form action='page2.php' method='post'>
First Name <input type='text' name='firstname'>
<br>
Last name <input type='text' name='lastname'>
<br><br>
<input type='submit' value='Submit'>
</form>
</body>
</html>
```

Submit Button

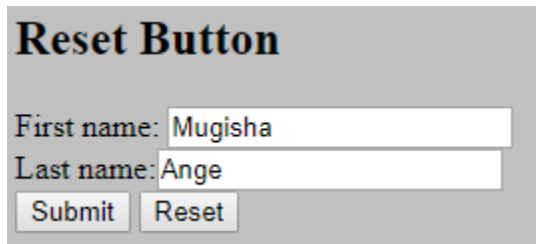
FirstName:

Last name:

Input: Reset

`<input type="reset">` defines a **reset button** that will reset all form values to their default values:

```
<form action="page2.php">
  First name: <input type="text" name="firstname" value="Mugisha"><br>
  Last name:<input type="text" name="lastname" value="Ange"><br>
  <input type="submit" value="Submit">
  <input type="reset">
</form>
```



Input Type Button

`<input type="button">` defines a **button**:

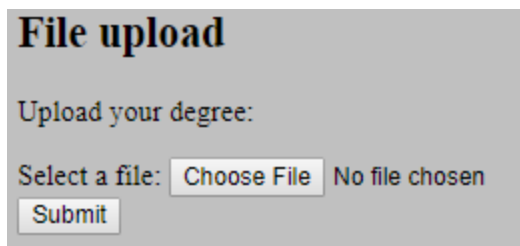
```
<input type="button" onclick="alert('Hello World!')" value="ClickMe!">
```

Input Type File

The `<input type="file">` defines a file-select field and a "Browse" button for file uploads.

```
<!DOCTYPE html>
<html>
<body bgcolor="silver">
<h2>File upload</h2>
<p>Upload your degree:</p>
<form action="page.php">
  Select a file: <input type="file" name="degree"><br>
  <input type="submit">
</form>

</body>
</html>
```



File upload

Upload your degree:

Select a file: No file chosen

Input Type Email

The `<input type="email">` is used for input fields that should contain an e-mail address.

Depending on browser support, the e-mail address can be automatically validated when submitted.

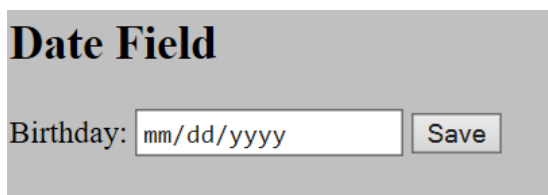
E-mail: `<input type="email" name="email">`

Input Type Date

The `<input type="date">` is used for input fields that should contain a date. Depending on browser support, a date picker can show up in the input field.

Birthday: `<input type="date" name="bday">`

Note: type="date" is not supported in Safari or Internet Explorer 11 and earlier versions



Date Field

Birthday:

Here is a list of some common input restrictions (some are new in HTML5):

Attribute	Description
disabled	Specifies that an input field should be disabled

Max(HTML5)	Specifies the maximum value for an input field
maxlength	Specifies the maximum number of character for an input field
Min(HTML5)	Specifies the minimum value for an input field
Pattern(HTML5)	Specifies a regular expression to check the input value against
Readonly	Specifies that an input field is read only (cannot be changed)
Required(HTML5)	Specifies that an input field is required (must be filled out)
Size	Specifies the width (in characters) of an input field
Step(HTML5)	Specifies the legal number intervals for an input field
Value	Specifies the default value for an input field

Show hint in HTML 5 text fields

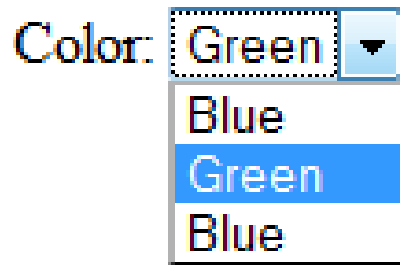
To have your text input field pre-filled in with some text that disappears as soon as the user selects it, use the attribute `placeholder`:

Username

Controls: Select

- ☐ These are used to allow the user to select one or more values from many
- ☐ Useful attributes:
 - **size**: the number of lines to display – when this is the default value of one, the control becomes a dropdown list
 - **multiple**: if this is set (`multiple="multiple"`) the user can select more than one value

```
<select name="color">
  <option value="red">Red</option>
  <option value="green">Green</option>
  <option value="blue">Blue</option>
</select>
```



Controls: Text Area

- ❑ This is a multi-line text input
- ❑ Initial text goes between the start and end tags, i.e. no value attribute
- ❑ Useful attributes:
 - **cols**: the width of the text area in characters
 - **rows**: the height of the text area in text lines

```
<!Doctype html>
<html>
<head>
<title> Forms</title>
</head>
<body bgcolor="silver">
<h3> Leave us a comment </h3>
<form action="#" method="get">
  FirstName<input type="text" name="firstName" /><br>

  Comment <textarea name="mytext" rows="3" cols="30">
  Put your Comment Here </textarea> <br>

  <input type="submit" value="Comment" />
</form>
</body>
</html>
```

HTML5 also has the following form input types:

- `<input type="url">`
- `<input type="week">`
- `<input type="time">`
- `<input type="tel">`
- `<input type="range">`
- `<input type="image">`
- `<input type="month">`
- `<input type="number">`
- `<input type="hidden">`
- `<input type="datetime-local">`
- `<input type="color">`

NB: New input types that are not supported by older web browsers, will behave as `<input type="text">`, `<select>`.

Others form elements includes

`<output>`
`<datalist>`
`<button>`

HTML pattern Attribute for `<input>`

The pattern attribute specifies a regular expression that the `<input>` element's value is checked against on form submission.

The pattern attribute works with the following input types: text, date, search, url, tel, email, and password.

Examples.

Code

```
<input type="text" name="username"
pattern="[A-Z]{6}" title="Use 6
uppercase letters">
```

The user is allowed to use 6 UPPERCASE letters only

```
<input type="text" name="username"
pattern="[A-Za-z]{6}" title="Use six
letters">
<input type="text" name="username"
pattern="[A-Za-z0-9]{6}" title="Use
alphanumeric">
<input type="text" name="username"
pattern="[a-z]{1,12}" title="Use only
no more than 12 letters">
```

```
<input type="password" name="password"
pattern=".{6,}" title="Six or more
characters">
```

The user is allowed to enter a username of 6 letters.

The user is allowed to specify the username which contains alphanumeric values.

The username should only consist of lowercase letters; no capital letters, numbers or other special characters allowed. In addition, the username length shouldn't be more than 12 characters.

The password should be six or more characters.

1.3.15 <address> Tag In Html

The <address> tag defines the contact information for the author/owner of a document or an article.

If the <address> element is inside the <body> element, it represents contact information for the document.

If the <address> element is inside an <article> element, it represents contact information for that article.

The text in the <address> element usually renders in *italic*. Most browsers will add a line break before and after the address element.

```
address {  
  display: block;  
  font-style: italic;  
}
```

Example: Reproduce and preview the following page in html

```
<!DOCTYPE html>  
<html>  
<head>  
  <title> Address tag</title>  
</head>  
<body>  
  
  <address>  
Write to <a href="mailto:info@rca.ac.rw">Rwanda Coding Academy</a>.<br>  
Visit us at:www.rca.ac.rw<br>  
Nyabihu District  
Western Province<br>  
Rwanda  
</address>  
  
</body>  
</html>
```

1.3.16 The <fieldset> tag in html

The <fieldset> tag is used to group related elements in a form. The <fieldset> tag draws a box around the related elements.

```

<!Doctype html>
<html lang="en">
<head>
<title> Fieldset and Form</title>
<meta charset="UTF-8">
</head>
<body style="background-color:silver">
<h3> Person identification form-Fieldset usage</h3>

<form action="#" method="get">
<fieldset><legend> Personal details</legend>
FirstName<input type="text" name="firstName" placeholder="Enter your firstName"><br>
LastName<input type="text" name="lastName"><br>
DoB<input type="date" name="dob"><br>
</fieldset>
<fieldset><legend> Address</legend>
District<input type="text" name="district" placeholder="Enter your district"><br>
Sector<input type="text" name="sector" placeholder="Enter your Sector"><br>
Street Number<input type="text" name="street" ><br>
</fieldset>
<fieldset><legend> Contact details</legend>

Telephone<input type="text" name="telephone" placeholder="Enter your phone"><br>
PO BOX<input type="text" name="pobox" placeholder="Enter your post office box "><br>
Email<input type="email" name="myemail"><br>
</fieldset>
<input type="submit" value="Submit details">
</form>
</body>
</html>

```

Results in browser

Person identification form-Fieldset usage

Personal details

FirstName Enter your firstName

LastName

DoB mm/dd/yyyy

Address

District Enter your district

Sector Enter your Sector

Street Number

Contact details

Telephone Enter your phone

PO BOX Enter your post office box

Email

Submit details

Below are fieldset attributes. Fieldset also supports Global attributes.

Attribute	Value	Description
<u>disabled</u>	disabled	Specifies that a group of related form elements should be disabled

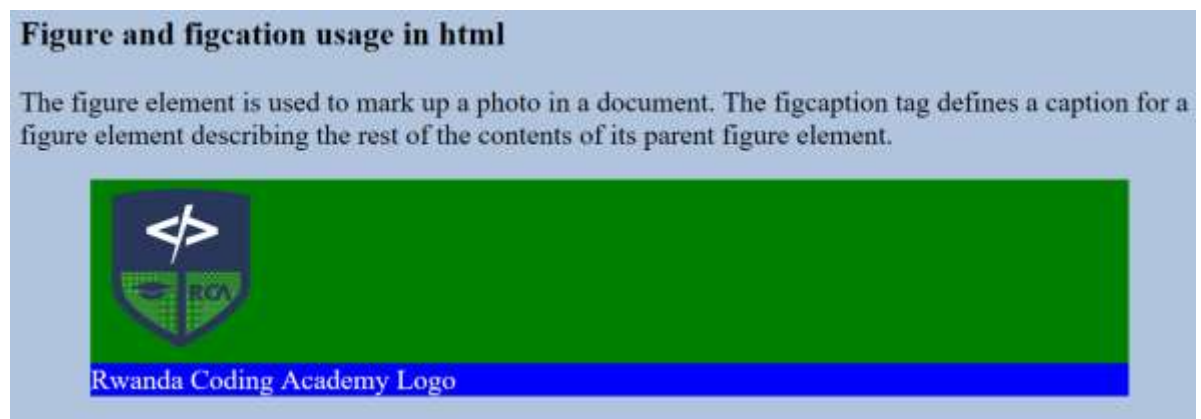
<u>form</u>	<i>form_id</i>	Specifies one or more forms the fieldset belongs to
<u>name</u>	<i>text</i>	Specifies a name for the fieldset

1.3.17 Figure and figcaption

The <figure> element is used to mark up a photo in a document. The <figcaption> tag defines a caption for a <figure> element. Example. Create an html document to reproduce the following codes.

```
<!doctype html>
<html lang="en">
<head>
<title> Figure and figcaption</title>
<meta charset="UTF-8">
<style>
img {
width: 100px;
height:100px;
}
figure{
background-color: green;
}
figcaption{
background-color: blue;
color:white;
}
</style>
</head>
<body style="background-color:lightsteelblue">
<h3> figure and figcaption usage in html</h3>
The figure element is used to mark up a photo in a document.
The figcaption tag defines a caption for a figure element describing the rest of the contents of its parent figure element.
<figure>
 Rwanda Coding Academy Logo</figcaption>
</figure>
</body>
</html>
```

Browser Result: The caption is added at the bottom of the image/figure.



1.3.18 The <section> tag in html

The <section> tag defines section in a document, such as chapters, headers, footers, or any other sections of the document.

- If you want to group elements to a **generic container**, use `<div>` tag instead.
- Elements which are inside one section are usually **related to each other**. Combined they make one unique element.
- Section element gives **meaning** to content, which is detectible by **search engines**.

Activity: Reproduce the following codes and preview them in a browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Caption usage in html table</title>

</head>
<body>
<section>
<h1>Volcano National Park</h1>
<p>The Volcano National Park lies along the Virunga Mountains,
with 8 ancient volcanoes, which are shared by Rwanda, Uganda and the Democratic Republic of Congo.
Just a short two hour drive from Rwanda's capital of Kigali, the park is a central
location for exploring some distinctly Rwandan experiences.
The visit to the mountain gorillas is often at the top of visitors.</p>

</section>

</body>
</html>
```

1.3.19 HTML `<label>` Tag

The `<label>` element does not render as anything special for the user. However, it provides a usability improvement for mouse users, because if the user clicks on the text within the `<label>` element, it toggles the control.

The `for` attribute of the `<label>` tag should be equal to the `id` attribute of the related element to bind them together.

Attribute	Value	Description
<u>for</u>	<i>element_id</i>	Specifies which form element a label is bound to
<u>form</u>	<i>form_id</i>	Specifies one or more forms the label belongs to

Example:

```
<h3> Authentication form</h3>
<form action="#" method="get">
  <label for="username">Username</label><input type="text" id="username" name="userName" ><br>
  <label for="pass">Password</label><input type="password" id="pass" name="mypassword"> <br>

  <input type="submit" value="LOGIN">
</form>
```

1.3.20 The HTML `<span>` element

The `<span>` tag is used to group inline-elements in a document.
The `<span>` tag provides no visual change by itself.

The `<span>` tag provides a way to add a hook to a part of a text or a part of a document.

Example

```
<div>The following span is an <span class="highlight">inline element</span>;  
its background has been colored to display both the beginning and end of  
the inline element's influence.</div>  
  
</body>  
</html>
```

The following span is an inline element; its background has been colored to display both the beginning and end of the inline element's influence.

1.5 HTML ENTITIES

Reserved characters in HTML must be replaced with character entities.

Characters that are not present on your keyboard can also be replaced by entities.

Some characters are reserved in HTML. If you use the less than (`<`) or greater than (`>`) signs in your text, the browser might mix them with tags.

Character entities are used to display reserved characters in HTML

To display a character entity:

`&entity_name;`
OR
`&#entity_number;`

[Click here to explore all html entities on W3C](#) or browse

<https://dev.w3.org/html5/html-author/charref>

Examples of Common used html entities

Result	Description	Entity Name	Entity Number
--------	-------------	-------------	---------------

	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	Ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark (apostrophe)	'	'
£	Pound	£	£
€	Euro	€	€
©	Copyright	©	©
®	registered trademark	®	®
≠	Not equal to	≠	≠
±	Plus/minus	±	±
°	Degree	°	°
≤ ≥	Less/greater than or equal to	≤ ≥	≤ ≥
½	Fractional half	½	¼
« »	Guillemets	« »	« »
≈	Almost Equal	≈	≈
✓	Check	✓ ✓	✓
⅓	Fraction: One third	⅓	⅓
⅕	Fraction :five sixth	⅚	⅚
→	Right arrow	→ → → → →	→ →
←	Left arrow	← ← ← ← ←	← ←
∞	Infini	∞	∞ ∞
№	Numero	№	№ № ;

Note: Entity names are case sensitive.

The following are used to mention card symbols

Sign	Description	Entity Name	Entity Number
♠	Spade	♠ ♠	♠ ♠
♣	Club	♣ ♣	♣ ♣
♥	Heart	♥ ♥	♥ ♥
♦	Diamond	♦ ♦	♦ ♦

1.6 Multimedia in html

1.6. 1 Videos in HTML

Before HTML5, a video could only be played in a browser with a plug-in (like flash).

The HTML5 `<video>` element specifies a standard way to embed a video in a web page.

The `controls` attribute adds video controls, like play, pause, and volume.

It is a good idea to always include `width` and `height` attributes. If height and width are not set, the page might flicker while the video loads.

The `<source>` element allows you to specify alternative video files which the browser may choose from. The browser will use the first recognized format.

The text between the `<video>` and `</video>` tags will only be displayed in browsers that do not support the `<video>` element.

```
<video width="320" height="240">
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
</video>
```

Using `<embed>` tag to add a video

```
<embed src="inside.mp4" autostart="false" height="300"
volume="60" width="144" />
```

Attributes for embed tag

- ❑ **autoplay** - sets whether the file will run automatically after the page is loaded. Can have the true or false value
- ❑ **hidden** - sets whether the buttons are hidden or not. Can have the true or false value.
- ❑ **volume** - can have any value from 0 to 100
- ❑ **loop** - sets whether the file will be replayed after it is finished. Can have the true or false value.
- ❑ **playcount** - this establishes how many times the file will be replayed. For example playcount="2" means it will be replayed two times and after it will stop.

HTML <video> Autoplay

To start a video automatically use the **autoplay** attribute

```
<video width="320" height="240" autoplay>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.ogg" type="video/ogg">  
</video>
```

MP4 is the new and upcoming format for internet video. MP4 is recommended by YouTube and is supported by HTML5.

Format	File	Description
Ogg	.ogg	Theora Ogg. Developed by the Xiph.Org Foundation. Supported by HTML5
WebM	.webm	WebM. Developed by the web giants, Mozilla, Opera, Adobe, and Google. HTML5.
MPEG-4 or MP4	.mp4	MP4. Developed by the Moving Pictures Expert Group. Based on QuickTime newer video cameras and TV hardware. Supported by all HTML5 browsers YouTube.

Playing a YouTube Video in HTML

As you might have to convert your videos to different formats to make them play in all browsers; converting videos to different formats can be difficult and time-consuming. An easier solution is to let YouTube play the videos in your web page.

YouTube will display an id (like tgbNymZ7vqY), when you save (or play) a video. You can use this id, and refer to your video in the HTML code.

To play your video on a web page, do the following:

- ☐ Upload the video to YouTube
- ☐ Take a note of the video id
- ☐ Define an <iframe> element in your web page
- ☐ Let the src attribute point to the video URL
- ☐ Use the width and height attributes to specify the dimension of the player
- ☐ Add any other parameters to the URL (see below)

<https://www.youtube.com/watch?v=ady7Ky0C4-g&t=455s>

```
<iframe width="400px" height="300px"
src="https://www.youtube.com/embed/ady7Ky0C4-g">
</iframe>
```

```
<iframe width="400px" height="300px"
src="https://www.youtube.com/embed/e9IXk00hmnI">
</iframe>
```

1.6.2 HTML Audio

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
</audio>
```

Tag	Description
<u><audio></u>	Defines sound content
<u><source></u>	Defines multiple media resources for media elements, such as <video> a

The `controls` attribute adds audio controls, like play, pause, and volume.

The `<source>` element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.

The text between the `<audio>` and `</audio>` tags will only be displayed in browsers that do not support the `<audio>` element.

Audio Formats

MP3 is the newest format for compressed recorded music. The term MP3 has become synonymous with digital music. If your website is about recorded music, MP3 is the choice.

Format	File	Description
WAV	.wav	WAV. Developed by IBM and Microsoft. Plays well on Windows, Macintosh, and other systems. Supported by HTML5.
Ogg	.ogg	Ogg. Developed by the Xiph.Org Foundation. Supported by HTML5.
MP3	.mp3	MP3 files are actually the sound part of MPEG files. MP3 is the most popular format for digital music players. Combines good compression (small files) with high quality. Supported by HTML5.

1.6.3 The HTML `<svg>` Element

- ☐ SVG stands for Scalable Vector Graphics
- ☐ SVG is used to define graphics for the Web
- ☐ SVG is a W3C recommendation

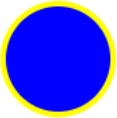
The HTML `<svg>` element is a container for SVG graphics. SVG has some predefined shape elements that can be used by developers:

- ☐ Rectangle `<rect>`
- ☐ Circle `<circle>`
- ☐ Ellipse `<ellipse>`
- ☐ Line `<line>`

- ☐ Polyline <polyline>
- ☐ Polygon <polygon>
- ☐ Path <path>

SVG has several methods for drawing paths, boxes, circles, text, and graphic images.

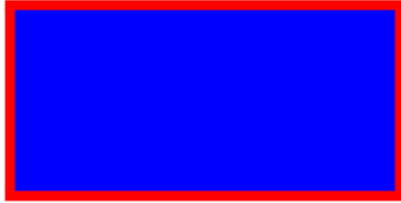
```
<!DOCTYPE html>
<html lang="en">
<meta charset="UTF-8">
<head>
<title>SVG</title>
</head>
<body>
  <svg width="100" height="100">
    <circle cx="50" cy="50" r="40"
      stroke="yellow" stroke-width="4" fill="blue"/>
  </svg>
</body>
</html>
```



- ☐ An SVG image begins with an <svg> element
- ☐ The width and height attributes of the <svg> element define the width and height of the SVG image
- ☐ The <circle> element is used to draw a circle
- ☐ The cx and cy attributes define the x and y coordinates of the center of the circle. If cx and cy are not set, the circle's center is set to (0, 0)
- ☐ The r attribute defines the radius of the circle
- ☐ The stroke and stroke-width attributes control how the outline of a shape appears. We set the outline of the circle to a 4px green "border"
- ☐ The fill attribute refers to the color inside the circle. We set the fill color to yellow
- ☐ The closing </svg> tag closes the SVG image

SVG Rectangle

```
<svg width="200" height="100">
  <rect width="200" height="100" style="fill:rgb(0,0,255);stroke-
width:10;stroke:rgb(255,0,0)" />
</svg>
```



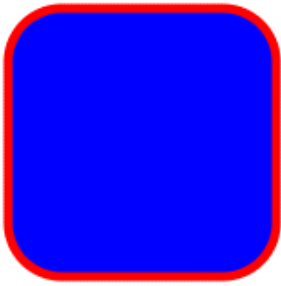
- ☐ The width and height attributes of the <rect> element define the height and the width of the rectangle
- ☐ The style attribute is used to define CSS properties for the rectangle
- ☐ The CSS fill property defines the fill color of the rectangle
- ☐ The CSS stroke-width property defines the width of the border of the rectangle
- ☐ The CSS stroke property defines the color of the border of the rectangle

```
<svg width="400" height="180">
  <rect x="50" y="20" width="150" height="100"
  style="fill:blue;stroke:pink;stroke-width:5;fill-opacity:0.1;stroke-
opacity:0.9" />
</svg>
```

- ☐ The x attribute defines the left position of the rectangle (e.g. x="50" places the rectangle 50 px from the left margin)
- ☐ The y attribute defines the top position of the rectangle (e.g. y="20" places the rectangle 20 px from the top margin)
- ☐ The CSS fill-opacity property defines the opacity of the fill color (legal range: 0 to 1)
- ☐ The CSS stroke-opacity property defines the opacity of the stroke color (legal range: 0 to 1)

SVG Rounded Rectangle

```
<svg width="400" height="180">
  <rect x="50" y="20" rx="20" ry="20" width="150" height="150"
  style="fill:blue;stroke:red;stroke-width:5;opacity:1" />
</svg>
```



SVG Ellipse - <ellipse>

```
<svg height="140" width="500">
  <ellipse cx="200" cy="80" rx="100" ry="50"
    style="fill:yellow;stroke:purple;stroke-width:2" />
</svg>
```

- ☐ The cx attribute defines the x coordinate of the center of the ellipse
- ☐ The cy attribute defines the y coordinate of the center of the ellipse
- ☐ The rx attribute defines the horizontal radius
- ☐ The ry attribute defines the vertical radius

SVG <line>

```
<svg height="210" width="500">
  <line x1="0" y1="0" x2="200" y2="200" style="stroke:rgb(255,0,0);stroke-
width:2" />
</svg>
```

- ☐ The x1 attribute defines the start of the line on the x-axis
- ☐ The y1 attribute defines the start of the line on the y-axis
- ☐ The x2 attribute defines the end of the line on the x-axis
- ☐ The y2 attribute defines the end of the line on the y-axis

SVG Polygon

```
<svg height="250" width="500">
<polygon points="30,40 100, 20 160,40 160,140 90, 140 85, 70"
style="fill:lime;stroke:purple;stroke-width:1"/>
</svg>
```



Polygon for SVG star

```
<svg width="300" height="200">
  <polygon points="100,10 40,198 190,78 10,78 160,198"
    style="fill:lime;stroke:purple;stroke-width:5;fill-rule:evenodd;" />
</svg>
```



SVG TEXT

```
<svg height="30" width="200">
  <text x="0" y="15" fill="red">Rwanda Coding Academy</text>
</svg>
```

```
<svg height="600" width="500">
  <text x="10" y="40" fill="blue" transform="rotate(30 20,40)" style="font-
size:52px"> Rwanda Coding Academy </text>
</svg>
```

More examples

1. https://www.w3schools.com/graphics/svg_examples.asp

Complex SVG online editor.

<https://svg-edit.github.io/svgedit/releases/latest/editor/svg-editor.html>

1.7 HTML Global Attributes

HTML attributes give elements meaning and context.

The global attributes below can be used on **any** HTML element.

HTML5= Attribute added in HTML5.

Attribute	Html Version	Description
accesskey		Specifies a shortcut key to activate/focus an element
class		Specifies one or more classnames for an element (refers to a class in a style sheet)
contenteditable	Html5	Specifies whether the content of an element is editable or not
data-*	Html5	Used to store custom data private to the page or application
dir		Specifies the text direction for the content in an element
draggable	Html5	Specifies whether an element is draggable or not
hidden	Html5	Specifies that an element is not yet, or is no longer, relevant
id		Specifies a unique id for an element
lang		Specifies the language of the element's content
spellcheck	Html5	Specifies whether the element is to have its spelling and grammar checked or not
style		Specifies an inline CSS style for an element
tabindex		Specifies the tabbing order of an element
title		Specifies extra information about an element

1. Accesskey Global attribute

The accesskey attribute specifies a shortcut key to activate/focus an element.

The **accesskey** global attribute provides a hint for generating a keyboard shortcut for the current element. The attribute value must consist of a single printable character (which includes accented and other characters that can be generated by the keyboard).

Note: The way of accessing the shortcut key is varying in different browsers:

Syntax:

`<element accesskey="character">`

Try the following codes

```
<!DOCTYPE html>
<html>
<head>
<title> Use of Accesskey attribute</title>
<style>
kbd{
background-color:black;
color:white;
}
</style>
</head>

<body>
<h2>Accesskey</h2>
<p>The accesskey attribute specifies a shortcut key to activate/focus an element.</p>

<ul>
<li>IE, Chrome, Safari, Opera 15+: <kbd>ALT</kbd> + <em>accesskey</em></li>
<li>Firefox: <kbd>ALT</kbd>+<kbd>SHIFT</kbd> + <em>accesskey</em></li>
</ul>
<h3> Example1. To open these links use shortcuts w and c  respectively:</h3>
<a href="https://www.w3schools.com/html" accesskey="t">HTML tutorial</a><br>
<a href="https://www.w3schools.com/css" accesskey="c">CSS tutorial</a>

<p><strong>Note:</strong> The shortcut is varying in different browsers and OS:</p>

<h3> Example2. To submit the form:</h3>
<form action="#" method="get">
<label for="email">Email</label><input type="email" name="myemail" id="email">
<input accesskey="h" type="Submit" value="Send">

</form>
<h3> Try</h3>
<kbd>alt</kbd> + <kbd>H</kbd> <br><b>OR</b><br>
<kbd>alt</kbd> + <kbd>shift</kbd> + <kbd>H</kbd>
</body>
</html>
```


Browsers and Operating System Support

	Windows	Linux	Mac
Firefox	Alt + Shift + key		On Firefox 57 or newer, Control + Option + key -OR- Control + Alt + key On Firefox 14 or newer, Control + Alt + key On Firefox 13 or older, Control + key
Internet Explorer	Alt + key	N/A	
Google Chrome	Alt + key		Control + Alt + key
Safari	Alt + key	N/A	Control + Alt + key
Opera 15+	Alt + key		Control + Alt + key
Opera 12	Shift + Esc opens a contents list which are accessible by accesskey, then, can choose an item by pressing key		

2. HTML class Attribute

The class attribute specifies one or more classnames for an element. The class attribute is mostly used to point to a class in a style sheet. However, it can also be used by a JavaScript to make changes to HTML elements with a specified class.

Value	Description
Classname	Specifies one or more class names for an element. To specify multiple classes, separate the class names with a space, e.g. <code></code>. This allows you to combine several CSS classes for one HTML element. Naming rules: ☐ Must begin with a letter A-Z or a-z ☐ Can be followed by: letters (A-Za-z), digits (0-9), hyphens ("-"), and underscores (" _")

EXAMPLE:

```

<!DOCTYPE html>
<html>
<head>
<title> Multiple class usage</title>
<style>
h1.intro {
    color: blue;
    text-align: center;
}

.important {
    background-color: yellow;
}
</style>
</head>
<body>

<h1 class="intro important">Header 1</h1>
<p>A paragraph.</p>

</body>
</html>

```

3. Contenteditable

The contenteditable attribute specifies whether the content of an element is editable or not.

Syntax:

```
<element contenteditable="true|false">
```

Attribute Values

Value	Description
True	Specifies that the element is editable
False	Specifies that the element is not editable

Example:

```
<p contenteditable="true">This is an editable paragraph.</p>
```

4. HTML data-* Attribute

The data-* attributes is used to store custom data private to the page or application.

The data-* attributes gives us the ability to embed custom data attributes on all HTML elements.

The stored (custom) data can then be used in the page's JavaScript to create a more engaging user experience (without any server-side database queries).

Syntax

```
<element data-*="somevalue">
```

Where somevalue specifies the value of the attribute (as a string)

The data-* attributes consist of two parts:

1. The attribute name should not contain any uppercase letters, and must be at least one character long after the prefix "data-"
2. The attribute value can be any string

```
<!DOCTYPE html>
<html>
<head>
<title> Use of data-* attribute</title>
<script>
function showDetails(animal) {
    var animalType = animal.getAttribute("data-animal-type");
    var result=document.getElementById("result").innerHTML="The " + animal.innerHTML + " is a " +
    animalType + ".";
}
</script>
</head>
<body>

<h1>Animal Species</h1>
<p>Click on a species to see what type it is:</p>

<ul>
    <li onclick="showDetails(this)" id="owl" data-animal-type="bird">Owl</li>
    <li onclick="showDetails(this)" id="salmon" data-animal-type="fish">Salmon</li>
    <li onclick="showDetails(this)" id="gorilla" data-animal-type="primate">Gorilla</li>
</ul>
<span id="result">Initial</span>
</body>
</html>

<ul>
    <li data-animal-type="bird">Owl</li>
    <li data-animal-type="fish">Salmon</li>
```

```
<li data-animal-type="spider">Tarantula</li>
</ul>
```

5. Dir attribute

The dir attribute specifies the text direction of the element's content.

Syntax

```
<element dir="ltr|rtl|auto">
```

Value	Description
ltr	Default. Left-to-right text direction
rtl	Right-to-left text direction
auto	Let the browser figure out the text direction, based on the content (only recommended if the text direction is unknown)

Example

```
<p dir="rtl">Write this text right-to-left!</p>
```

6. Draggable

The draggable attribute specifies whether an element is draggable or not. Links and images are draggable by default. The draggable attribute is often used in drag and drop operations

```
<p draggable="true">This is a draggable paragraph.</p>
```

Note: More about draggable in coming API section

Syntax:

```
<element draggable="true|false|auto">
```

Value	Description
True	Specifies that the element is draggable
False	Specifies that the element is not draggable
Auto	Uses the default behavior of the browser

7. HTML hidden Attribute

SYNTAX

```
<element hidden>
```

The hidden attribute is a boolean attribute. When present, it specifies that an element is not yet, or is no longer, relevant. Browsers should not display elements that have the hidden attribute specified.

The hidden attribute can also be used to keep a user from seeing an element until some other condition has been met (like selecting a checkbox, etc.). Then, a JavaScript could remove the hidden attribute, and make the element visible.

Example:

`<p hidden>`This paragraph should be hidden.`</p>`

8. HTML id Attribute

The id attribute specifies a unique id for an HTML element (the value must be unique within the HTML document). The id attribute is most used to point to a style in a style sheet, and by JavaScript to manipulate the element with the specific id. It can be used also to link parts of a webpage.

Syntax

`<element id="id">`

Value	Description
<i>Id</i>	Specifies a unique id for the element. Naming rules: ☐ Must start with a letter ☐ Must contain at least one character ☐ Must not contain any space characters

Example:

```
<!Doctype html>
<html>
<head>
<title> Link parts of a webpage using ID</title>
</head>
<body>
<h2><a id="top">Some heading</a></h2>
<p>Lots of text....</p>
<p>Lots of text....</p>
<p>Lots of text....</p>
<a href="#top">Go to top</a>
</body>
</html>
```

9. HTML lang Attribute

The lang attribute specifies the language of the element's content.

Syntax

`<element lang="language_code">`

Value	Description
<i>language_code</i>	Specifies the language code for the element's content. More about language code on : https://www.w3schools.com/tags/att_global_lang.asp

Examples:

```
<p lang="rw">Iki gika kiri mu Kinyarwanda</p>
<p lang="fr">Ce Paragraph est en Français </p>
<p lang="en">This paragraph is in English</p>
<html lang="rw">
```

10. HTML spellcheck Attribute

Example:

```
<p contenteditable="true" spellcheck="true"> This is a praggagraph. It is editable. Try to change the text </p>
```

The spellcheck attribute specifies whether the element is to have its spelling and grammar checked or not.

The following can be spellchecked:

- ☐ Text values in input elements (not password)
- ☐ Text in <textarea> elements
- ☐ Text in editable elements

Syntax

`<element spellcheck="true|false">`

HTML style Attribute

The style attribute specifies an inline style for an element. The style attribute will override any style set globally, e.g. styles specified in the <style> tag or in an external style sheet.

Syntax

`<element style="style_definitions">`

Value	Description
<i>style_definitions</i>	One or more CSS properties and values separated by semicolons (e.g. style="color:blue;text-align:center")

Example:

`<h1 style="color:blue;text-align:center">This is a header</h1>`

More examples about style , refer to css notes

11. Tabindex

The tabindex attribute specifies the tab order of an element (when the "tab" button is used for navigating).

Syntax

`<element tabindex="number">`

Value	Description
<i>number</i>	Specifies the tabbing order of the element (1 is first)

`<a href="https://www.w3schools.com/" tabindex="2">W3Schools</a>`

`<a href="http://www.google.com/" tabindex="1">Google</a>`

`<a href="http://www.microsoft.com/" tabindex="3">Microsoft</a>`

12. HTML title Attribute

The title attribute specifies extra information about an element. The information is most often shown as a tooltip text when the mouse moves over the element.

Syntax

```
<element title="text">
```

```
<abbr title="Rwanda Coding Academy">RCA</abbr>
```

EVENT ATTRIBUTES

Global event attributes can be added to HTML elements to define event actions using Javascript

Reference: https://www.w3schools.com/tags/ref_eventattributes.asp

All attributes

https://www.w3schools.com/tags/ref_attributes.asp